Do Now · retrieval
Character sets: ASCII & Unicode

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Which data type should store a telephone number beginning with 0? Explain briefly.

Q2

Convert 41_{16} to denary and eight-bit binary.

Q3

Apply AND 00001111 to 10100101. Explain which bits the mask preserves.

Q4

In 6+4 two’s-complement floating-point format, subtract 010000 0010 from 011000 0010. Give a normalised result.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Characters need agreed codes — A character set defines characters and their assigned numeric codes. An encoding specifies how those codes are represented as bytes.

Core idea 2

Standard ASCII is limited — Seven bits provide $2^7 = 128$ codes for letters, digits, punctuation and control characters. — Eight-bit extensions can provide 256 values, but different extensions may map the extra codes differently.

Core idea 3

Unicode supports a much larger repertoire — Unicode assigns code points to characters from many writing systems and symbols. Its first 128 code points match ASCII.

Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Core idea 4

Use the encoding stated in storage questions — Storage depends on the number of code points and the number of bytes used for each one. Do not assume every Unicode character uses the same width. — CODE:
'0' in ASCII = 48 = 00110000_2
integer 0 = 0

Vocabulary · use the correct technical terms

Character set — A collection of characters with assigned codes.

Code point — A numeric identifier assigned to a character in Unicode.

Encoding — Rules for representing character codes as bytes.

ASCII — A standard seven-bit character code with 128 possible values.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Encode the text “AB”

Follow the worked method

1. Start with the character codes — CODE:
A = 65
B = 66 — ASCII assigns a numeric code to each character.

2. Write the 7-bit ASCII values — CODE:
A = 1000001
B = 1000010

3. Add a leading zero — CODE:
A → 01000001
B → 01000010

AB → 01000001 01000010 — The leading zero lets each 7-bit ASCII code sit inside an 8-bit byte; it does not expand the standard ASCII repertoire.

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Build the answer

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Question 1 · ADAPTED OCR PROGRAMMING TASK · [3]
Complete the three gaps to count uppercase ASCII letters. asc returns a character's code. A = 65 and Z = 90. The loop examines each character once.
CODE:

```
function countCapitals(text)
count = 0
for each character in text
code = asc(character)
if code >= 65 _____ code <= 90 then
count = _____
endif
next character
return _____
endfunction
```

Adapted from OCR June 2022 Q5(e)(i): simplified loop and changed gaps; classroom allocation of one mark per gap. OCR June 2022, Q5(e), page 24

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 2

Question 5 · ORIGINAL ENCODING APPLICATION · [4]
A UTF-8 string contains 20 ASCII code points, three euro-sign code points and two emoji code points. The byte widths are 1, 3 and 4 respectively. Ignore metadata. Calculate bytes [2] and bits [1]. Explain why multiplying the number of code points by 2 bytes is not valid here. [1]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Standard ASCII provides 128 character codes.

Use key vocabulary from memory

1. Explain Character set in your own words.

2. Explain Code point in your own words.

3. Explain Encoding in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.